

Assignment 1



Indian Institute of Technology
Hyderabad

VLSI Design

EE3301

Harshil Rathan Y

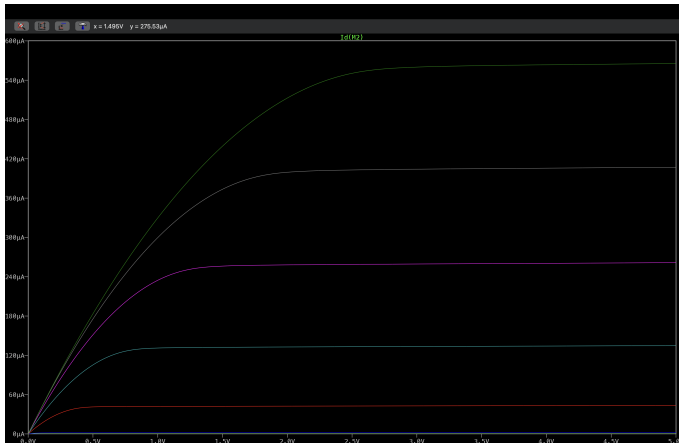
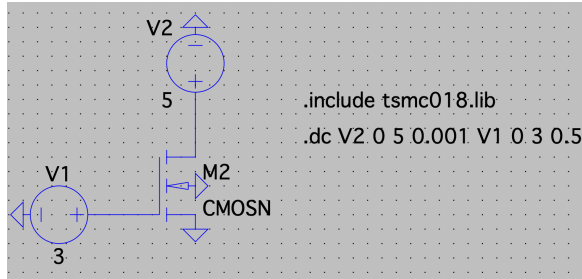
EE24BTECH11064

भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

1 I-V characteristics of an NMOS and PMOS

1.1 NMOS

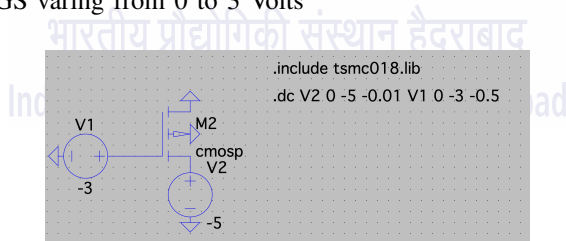
For different V_{GS} varying from 0 to 3 Volts

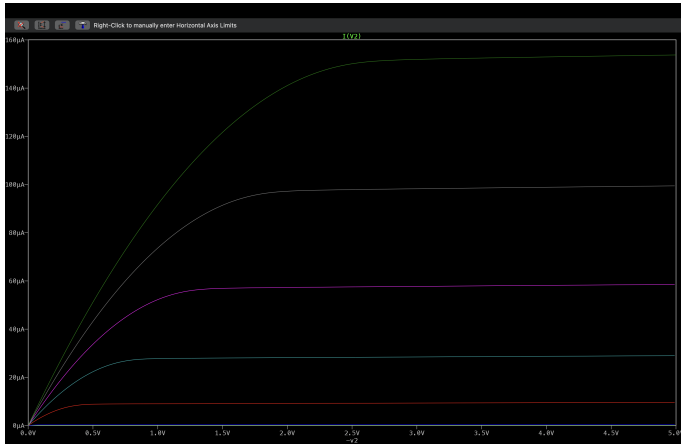


The NMOS drain characteristics were obtained successfully, showing that the drain current increases with V_{GS} and exhibits the expected transition from the linear region to saturation.

1.2 PMOS

For different V_{GS} varying from 0 to 3 Volts





The IV characteristics of both NMOS and PMOS transistors were successfully obtained for different gate to source voltages

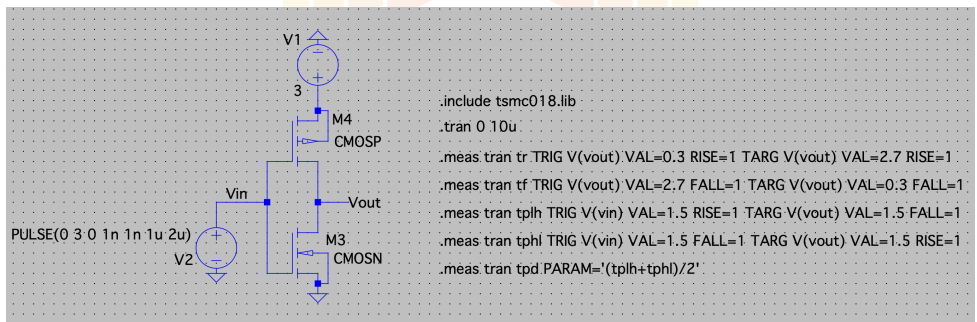
- For the NMOS, the drain current increases with increasing V_{GS} , and the output characteristics clearly demonstrate the transition from the linear region to the saturation region as V_{DS} increases. Higher values of V_{GS} result in higher drain currents.
- For the PMOS, the drain current magnitude increases with increasing negative V_{GS} . The characteristics similarly exhibit the expected transition from the linear region to the saturation region as the magnitude of V_{DS} increases.

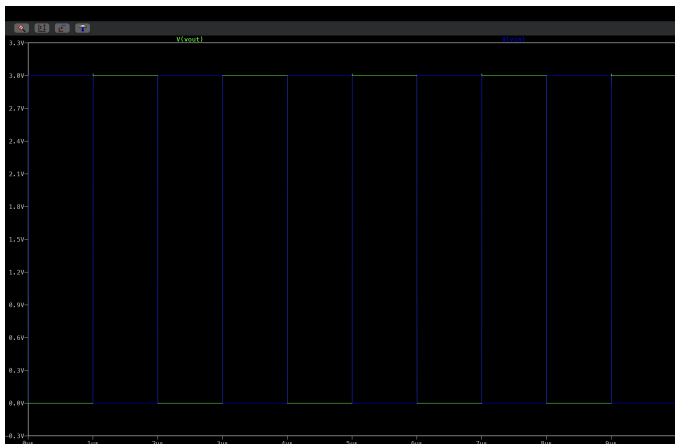
2 CMOS Inverter

When $V_{in} = 0V$, the PMOS turns ON and the NMOS turns OFF, charging the output to approximately V_{DD} . When $V_{in} = 3V$, the PMOS turns OFF and the NMOS turns ON, pulling the output approximately to ground. Thus, the circuit performs the required logic inversion.

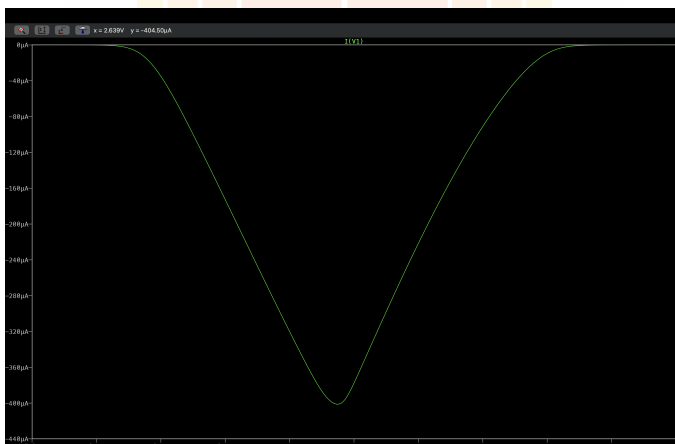
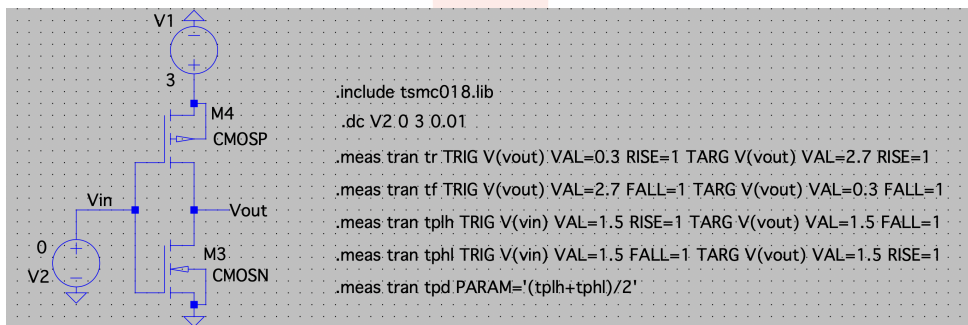
2.1 Without Capacitor

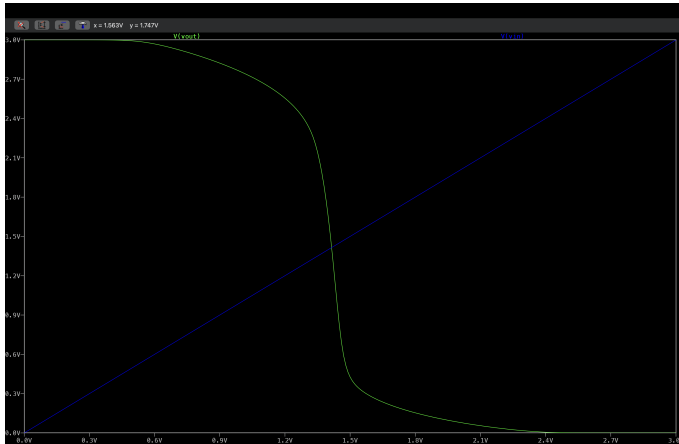
2.1.1 PULSE





2.1.2 DC Sweep





Parameter	Value
W_N	$1\ \mu\text{m}$
W_P	$1.9\ \mu\text{m}$
$L_N = L_P$	$0.18\ \mu\text{m}$
V_{DD}	3 V
Rise time t_r	0.420 ns
Fall time t_f	0.413 ns

```

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 27
method = modified trap
CompressWinPoints = 1024
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

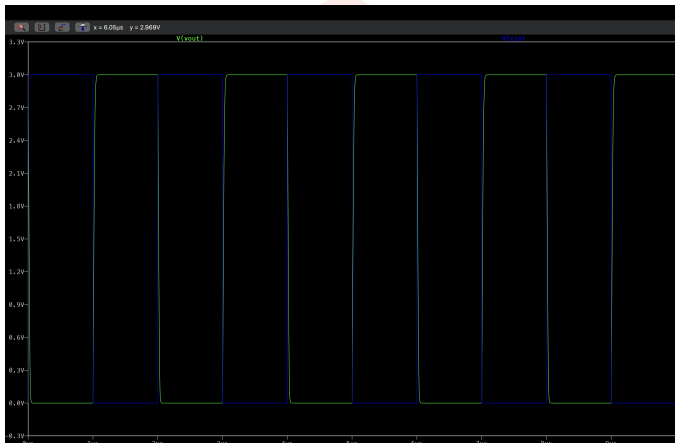
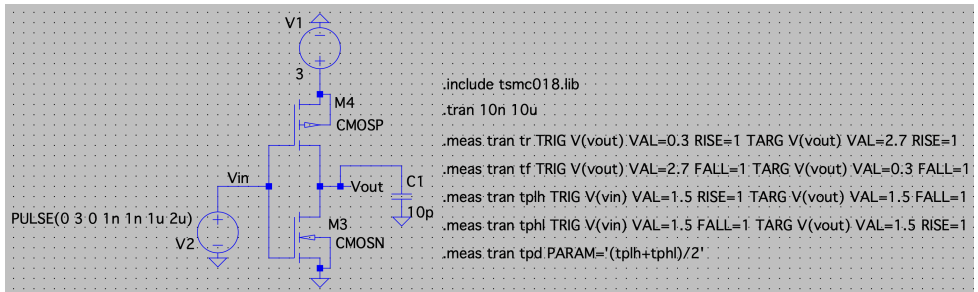
tr=4.20399e-10 FROM 1.00131e-06 TO 1.00173e-06
tf=4.12566e-10 FROM 2.52317e-10 TO 6.64883e-10
tplh=-2.17925e-11 FROM 5e-10 TO 4.78207e-10
tphl=6.31918e-11 FROM 1.0015e-06 TO 1.00156e-06
tpd: ((tplh+tphl)/2)=2.06996e-11

```

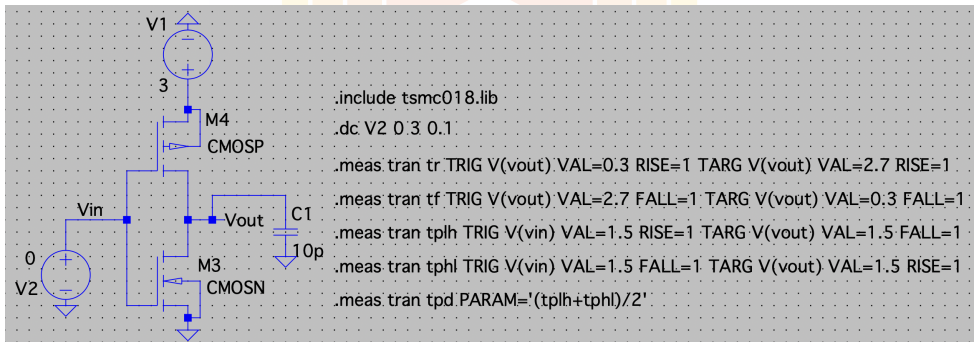
The rise and fall times are very close, indicating that the inverter has been appropriately sized for approximately symmetrical switching.

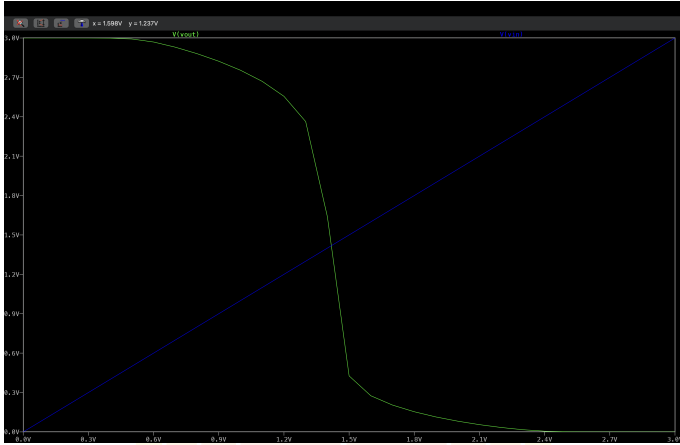
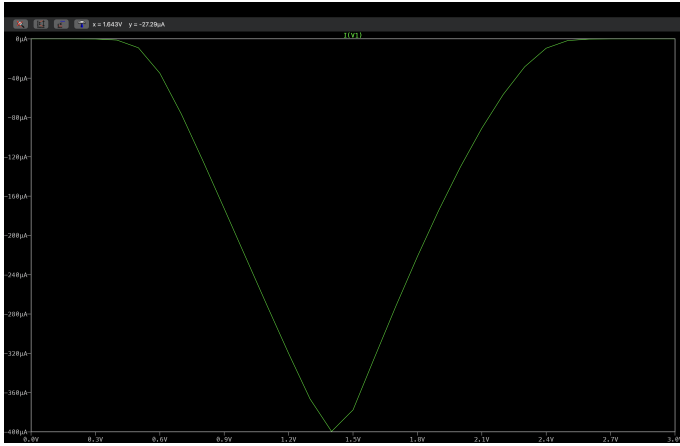
2.2 With Capacitor

2.2.1 PULSE



2.2.2 DC Sweep





Parameter	Value
W_N	1 μm
W_P	2 μm
$L_N = L_P$	0.18 μm
V_{DD}	3 V
Load capacitance C_L	10 pF
Rise time t_r	28.886 ns
Fall time t_f	26.480 ns

```

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 27
method = modified trap
CompressWinPoints = 1024
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr=2.88855e-08 FROM 1.00392e-06 TO 1.0328e-06
tf=2.648e-08 FROM 3.22501e-09 TO 2.9705e-08
tplh=1.40431e-08 FROM 5e-10 TO 1.45431e-08
tphl=1.2715e-08 FROM 1.0015e-06 TO 1.01422e-06
tpd: ((tplh+tphl)/2)=1.33791e-08

```

The addition of the 10 pF load capacitance significantly increases the rise time, fall time, and propagation delay of the CMOS inverter due to the increased capacitive charging and discharging requirement.

3 CMOS NAND, NOR Gates

3.1 NAND Gate

In CMOS implementation, the NAND gate consists of two PMOS transistors connected in parallel in the pull-up network and two NMOS transistors connected in series in the pull-down network .

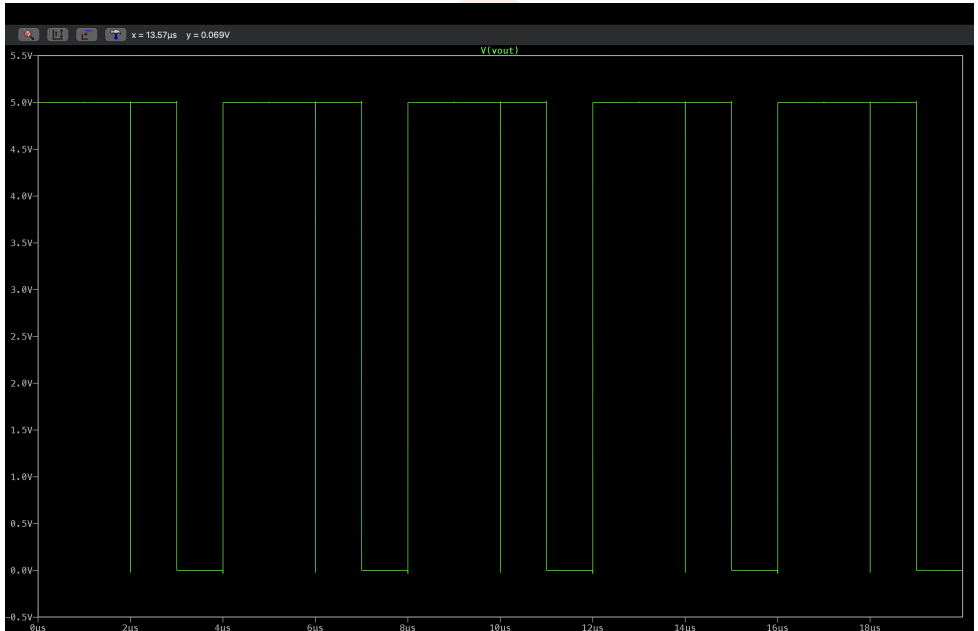
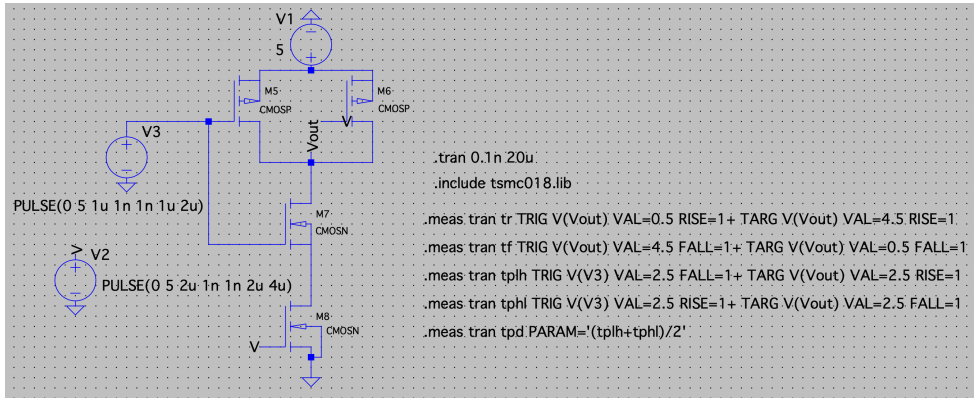
- When both A and B are HIGH, both NMOS transistors turn ON, creating a path from the output to ground. The PMOS transistors are OFF, giving $Y = 0$.
- When either input is LOW, at least one PMOS transistor turns ON and the corresponding NMOS series path is broken. The output is therefore pulled HIGH, giving $Y = 1$.

The truth table is

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The PMOS and NMOS widths can be adjusted to obtain approximately equal rise and fall times

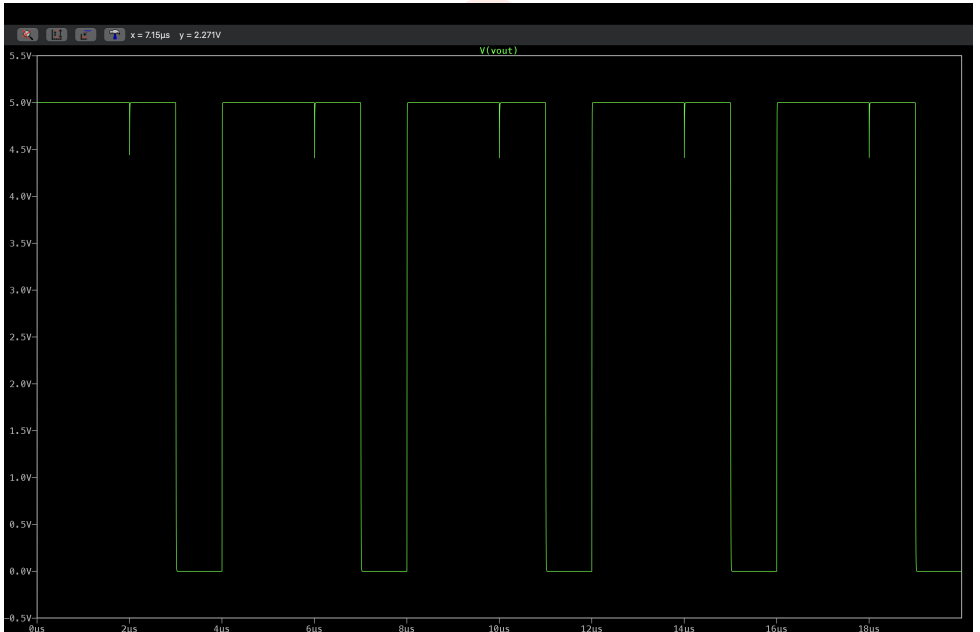
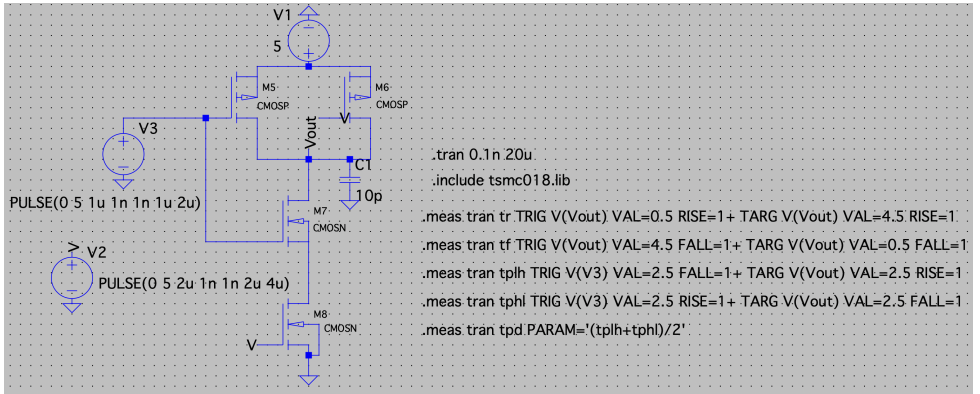
3.1.1 Without Capacitor



tr=3.0597939841e-10 FROM 2.00147938218e-06 TO 2.00178536158e-06
 tf=2.98267015434e-10 FROM 2.00017569969e-06 TO 2.0004739667e-06
 tphi=1.56824872879e-10 FROM 2.00150000001e-06 TO 2.00165682488e-06
 tphi=9.9985884652e-07 FROM 1.00049999999e-06 TO 2.0003588465e-06
 tpd: '(tphi+tphi)/2'=5.00007835696e-07

$$W_p = 2.1\mu m, \quad L_p = 0.18\mu m \quad W_n = 2\mu m, \quad L_n = 0.18\mu m$$

3.1.2 With Capacitor



```

tr=1.32812260975e-08 FROM 4.00258725551e-06 TO 4.01586848161e-06
tf=1.38462201568e-08 FROM 3.00173872661e-06 TO 3.01558494677e-06
tphi=2.00551848562e-06 FROM 2.00150000001e-06 TO 4.00701848563e-06
tphi=2.00604425578e-06 FROM 1.0005e-06 TO 3.00654425579e-06
tpd: '(tphi+tphi)/2'=2.0057813707e-06

```

$$W_p = 2.1\mu m, \quad L_p = 0.18\mu m \quad W_n = 2\mu m, \quad L_n = 0.18\mu m$$

3.2 NOR Gate

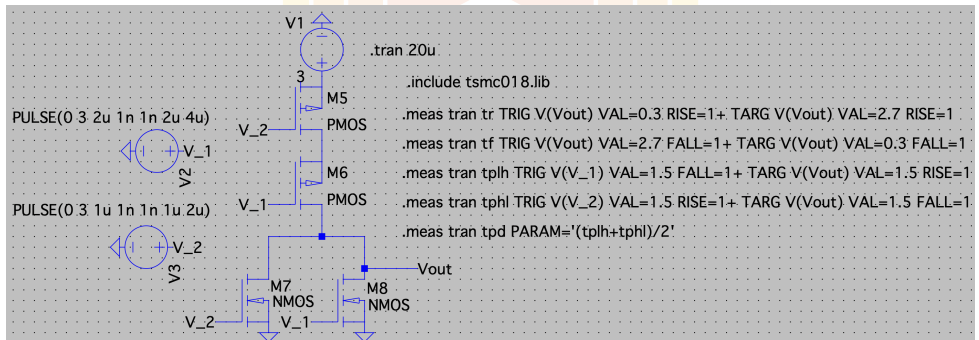
The CMOS implementation consists of two PMOS transistors connected in series in the pull-up network and two NMOS transistors connected in parallel in the pull-down network

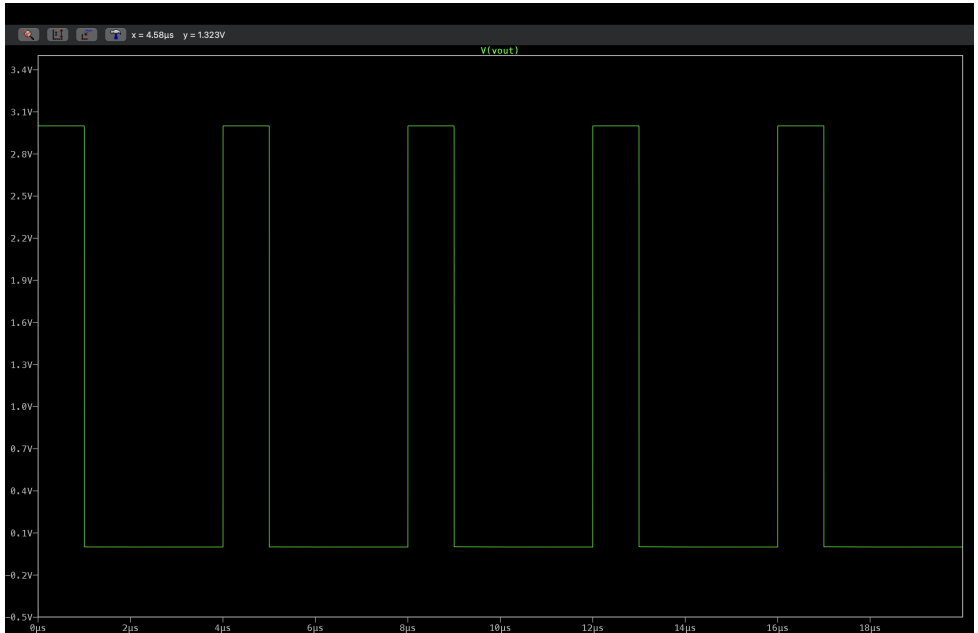
- When both A and B are LOW, both PMOS transistors turn ON, providing a path from V_{DD} to the output, while both NMOS transistors are OFF. Therefore, $Y = 1$.
- When either A or B is HIGH, the corresponding NMOS transistor turns ON and pulls the output to ground. At the same time, the corresponding PMOS transistor turns OFF, resulting in $Y = 0$.
- When both inputs are HIGH, both NMOS transistors are ON and both PMOS transistors are OFF, giving $Y = 0$.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Since the PMOS transistors in a NOR gate are connected in series, the pull-up network has a higher effective resistance than that of a NAND gate. Therefore, the PMOS widths are generally increased to improve the charging speed of the output and to obtain more balanced rise and fall times.

3.2.1 Without Capacitor





```

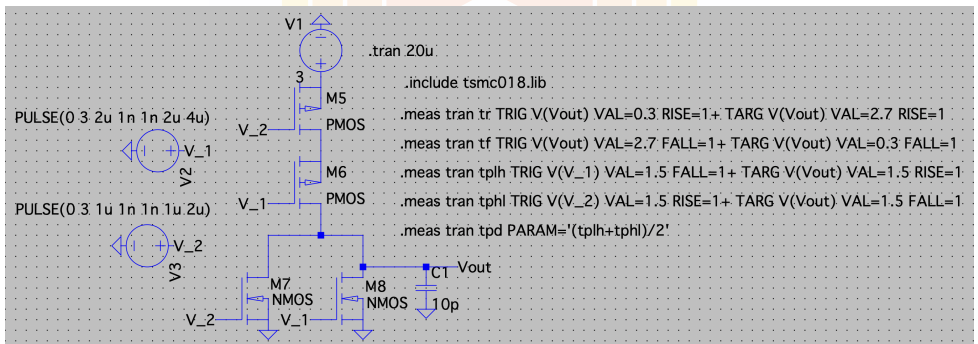
tr=3.73335e-10 FROM 4.00132e-06 TO 4.00169e-06
tf=3.91124e-10 FROM 1.00042e-06 TO 1.00081e-06
tph=6.38819e-12 FROM 4.0015e-06 TO 4.00151e-06
tphl=1.5102e-10 FROM 1.0005e-06 TO 1.00065e-06
tpd: ((tph+tphl)/2)=7.87042e-11

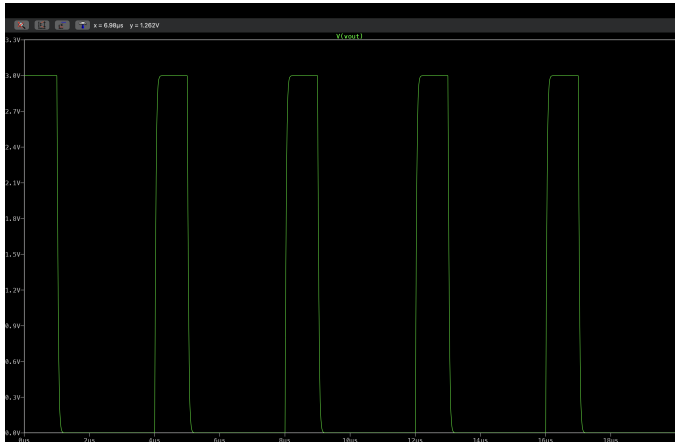
```

$$W_p = 4\mu m, \quad L_p = 0.18\mu m$$

$$W_n = 1.15\mu m, \quad L_n = 0.18\mu m$$

3.2.2 With Capacitor





```
tr=7.03656e-08 FROM 4.00651e-06 TO 4.07687e-06
tf=7.2396e-08 FROM 1.00587e-06 TO 1.07827e-06
tplh=2.82475e-08 FROM 4.0015e-06 TO 4.02975e-06
tphl=3.11649e-08 FROM 1.0005e-06 TO 1.03166e-06
tpd: ((tplh+tphl)/2)=2.97062e-08
```

$$W_p = 2.5\mu\text{m}, \quad L_p = 0.18\mu\text{m}$$

$$W_n = 1.15\mu\text{m}, \quad L_n = 0.18\mu\text{m}$$

4 3 Input NAND NOR

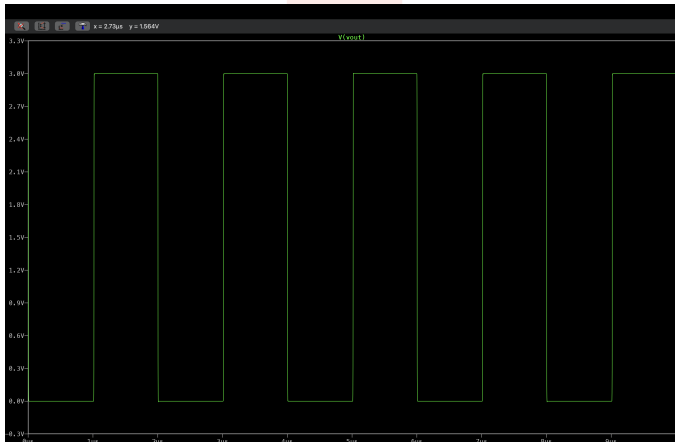
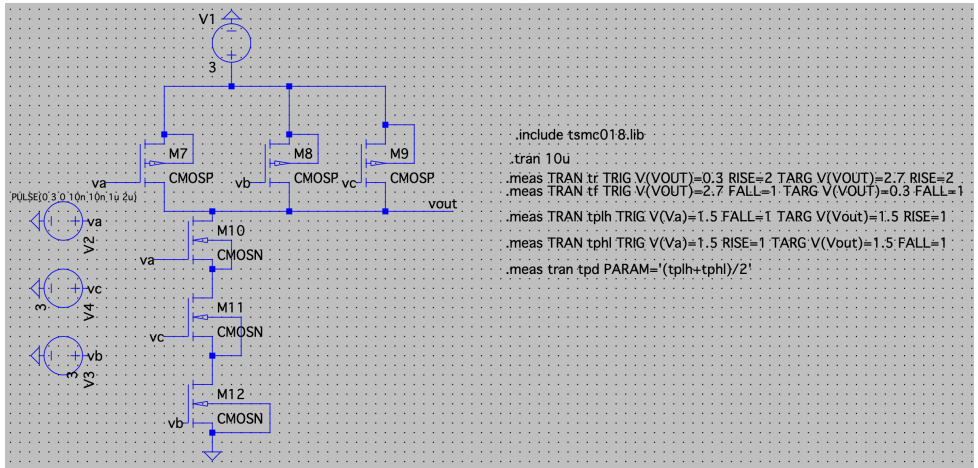
4.1 NAND

The CMOS implementation consists of three PMOS transistors connected in parallel in the pull-up network and three NMOS transistors connected in series in the pull-down network

- When A , B , and C are all HIGH, all three NMOS transistors turn ON, creating a conducting path from the output to ground. The PMOS transistors are OFF, resulting in $Y = 0$.
- When any one of the inputs is LOW, the corresponding NMOS transistor turns OFF, breaking the series pull-down path. At least one PMOS transistor turns ON and pulls the output HIGH, resulting in $Y = 1$.

Since the three NMOS transistors are connected in series, the effective resistance of the pull-down network is higher than that of a 2-input NAND gate. Therefore, the NMOS widths may be increased to improve the discharge speed and obtain more balanced rise and fall times.

4.1.1 Without Capacitor



```

tr=2.44776e-09 FROM 3.01465e-06 TO 3.01709e-06
tf=2.31872e-09 FROM 2.68645e-09 TO 5.00518e-09
tphi=1.03946e-09 FROM 1.015e-06 TO 1.01604e-06
tphi=-1.03007e-09 FROM 5e-09 TO 3.96993e-09
tpd: ((tphi+tphi)/2)=4.69106e-12

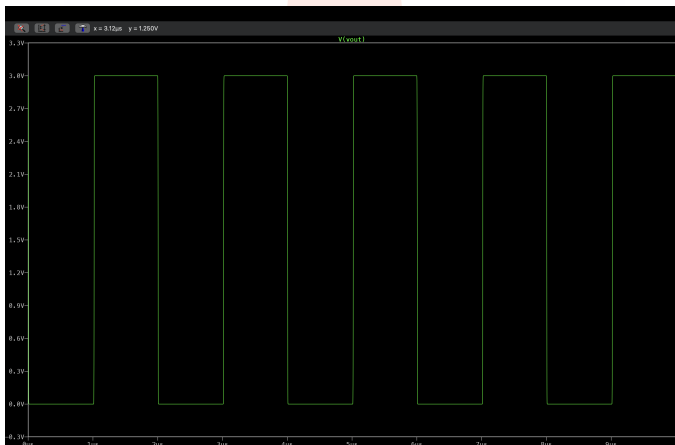
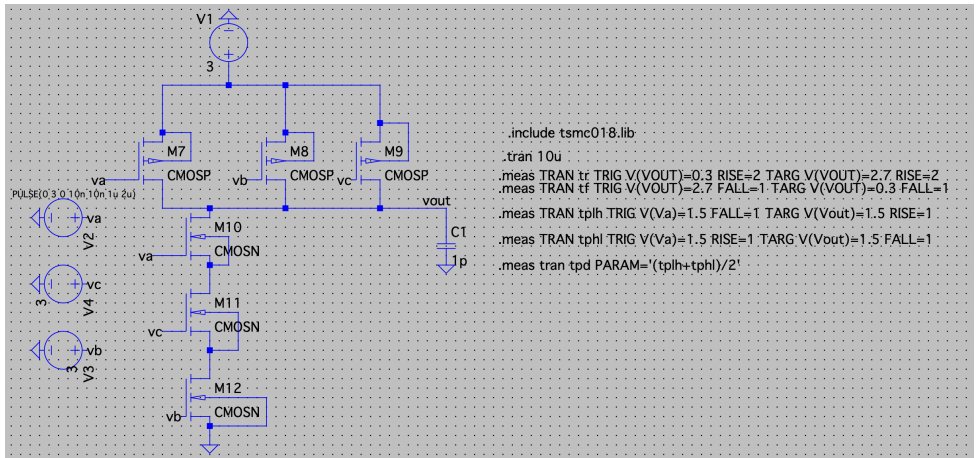
```

भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

$$W_p = 2\mu\text{m}, \quad L_p = 0.18\mu\text{m}$$

$$W_n = 3\mu\text{m}, \quad L_n = 0.18\mu\text{m}$$

4.1.2 With Capacitor



```

tr=4.53482e-09 FROM 3.0155e-06 TO 3.0203e-06
tf=4.58424e-09 FROM 3.35823e-09 TO 7.94247e-09
tphi=3.1548e-09 FROM 1.015e-06 TO 1.01815e-06
tphi=9.75041e-10 FROM 5e-09 TO 5.97504e-09
tpd: ((tphi+tphi)/2)=2.06492e-09

```

$$W_p = 2\mu\text{m}, \quad L_p = 0.18\mu\text{m}$$

$$W_n = 3\mu\text{m}, \quad L_n = 0.18\mu\text{m}$$

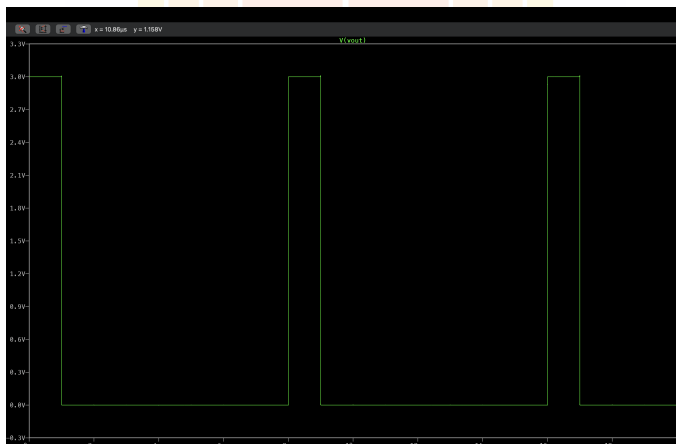
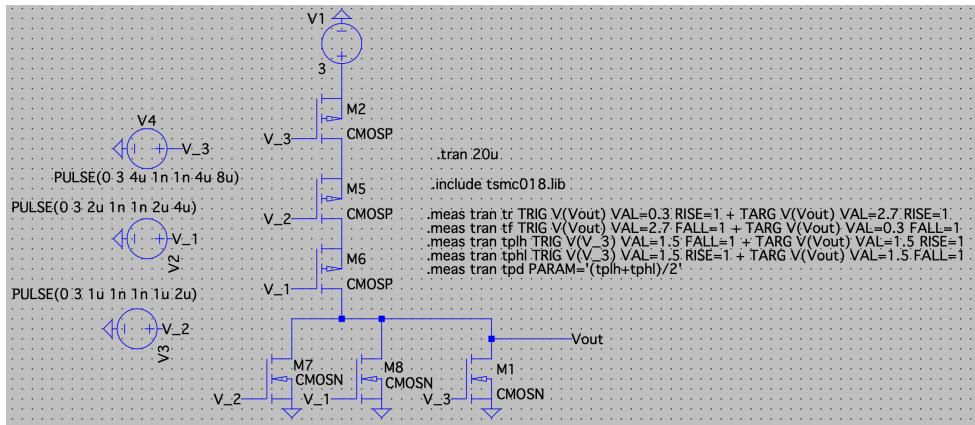
4.2 NOR

The CMOS implementation consists of three PMOS transistors connected in series in the pull-up network and three NMOS transistors connected in parallel in the pull-down

network

- When A , B , and C are all LOW, all three PMOS transistors turn ON, providing a path from V_{DD} to the output. The NMOS transistors remain OFF, resulting in $Y = 1$.
- When any one of the inputs is HIGH, the corresponding NMOS transistor turns ON and provides a path to ground. At the same time, the corresponding PMOS transistor turns OFF, pulling the output LOW.
- When all three inputs are HIGH, all NMOS transistors are ON and all PMOS transistors are OFF, resulting in $Y = 0$.

4.2.1 Without Capacitor



Indian Institute of Technology Hyderabad

```

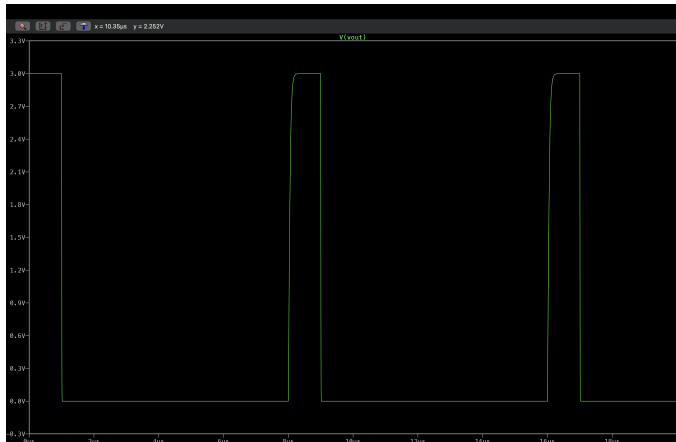
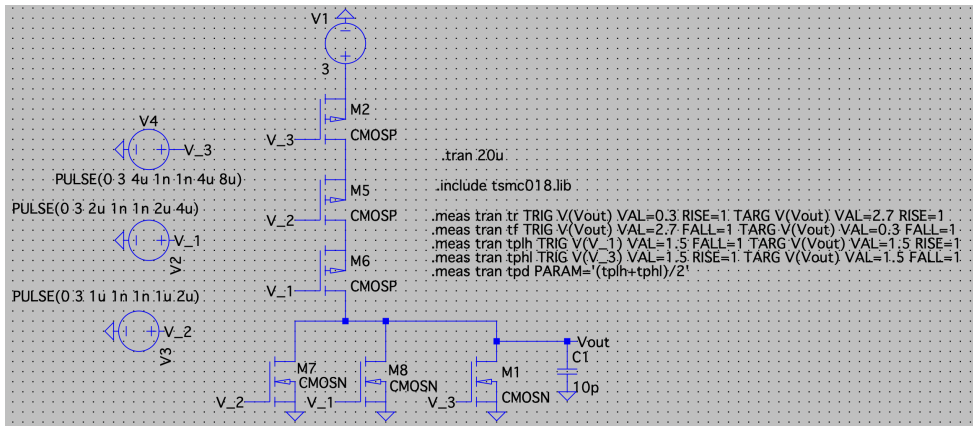
tr=1.62762e-10 FROM 8.00179e-06 TO 8.00195e-06
tf=1.77804e-10 FROM 1.00019e-06 TO 1.00037e-06
tphi=4.00036e-06 FROM 4.0015e-06 TO 8.00186e-06
tphi=-3.0002e-06 FROM 4.0005e-06 TO 1.0003e-06
tpd: ((tphi+tphi)/2)=5.00077e-07

```

$$W_p = 6.1\mu m, \quad L_p = 0.18\mu m$$

$$W_n = 1.05\mu m, \quad L_n = 0.18\mu m$$

4.2.2 With Capacitor



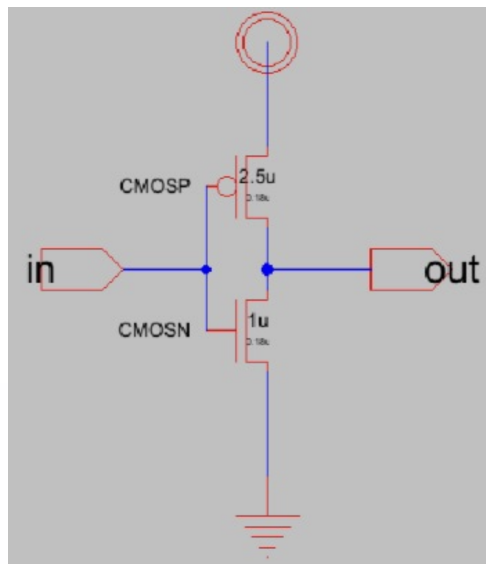
```
tr=2.46884e-08 FROM 8.00338e-06 TO 8.02806e-06
tf=2.47664e-08 FROM 1.00307e-06 TO 1.02784e-06
tph=4.00929e-06 FROM 4.0015e-06 TO 8.01079e-06
tphl=-2.98718e-06 FROM 4.0005e-06 TO 1.01332e-06
tpd: ((tph+tphl)/2)=5.11055e-07
```

$$W_p = 6.1\mu m, \quad L_p = 0.18\mu m$$

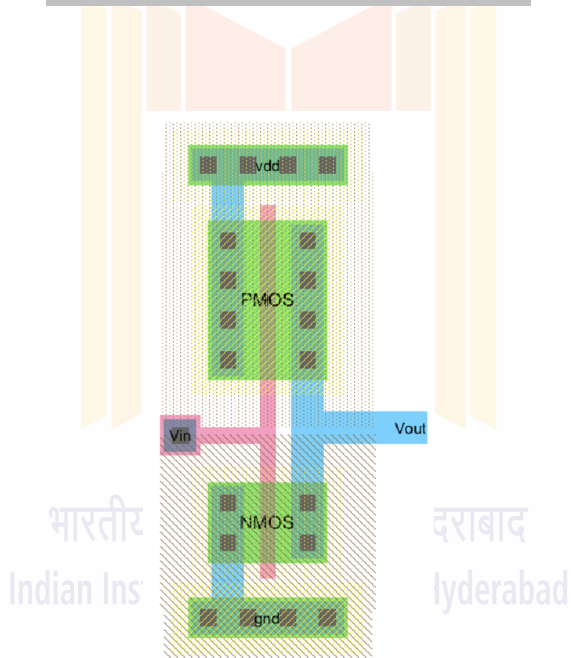
$$W_n = 1.05\mu m, \quad L_n = 0.18\mu m$$

5 CMOS Inverter Layout

The Circuit schematic for the CMOS inverter in Electric VLSI Design Tool



Layout



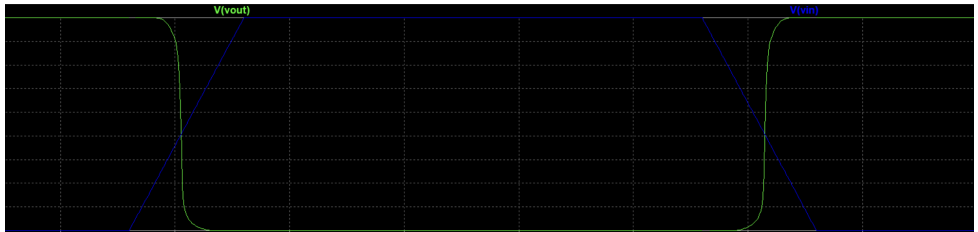
Antenna Errors

```
No antenna errors found (took 0.005 secs)
```

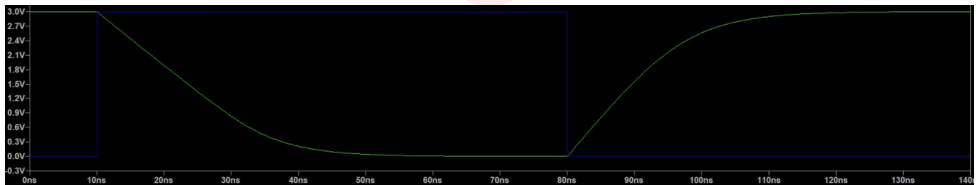
DRC Check

```
Running DRC with area bit on, extension bit on, Mosis bit
Checking again hierarchy .... (0.0 secs)
Found 7 networks
Checking cell 'Inv{lay}'
      No errors/warnings found
0 errors and 0 warnings found (took 0.006 secs)
```

Output



RC Extraction



Maximum Frequency Without Capacitor

```
tf=2.49272e-008 FROM 1.26571e-008 TO 3.75843e-008
tr=2.08773e-008 FROM 8.18455e-008 TO 1.02723e-007
tphl=1.35094e-008 FROM 1.0005e-008 TO 2.35144e-008
tplh=9.62857e-009 FROM 8.0015e-008 TO 8.96436e-008
```

$$F_{max} \approx 86.4MHz$$

Maximum Frequency With Capacitor

```
tf=2.55079e-008 FROM 1.2732e-008 TO 3.82399e-008
tr=2.13761e-008 FROM 8.18905e-008 TO 1.03267e-007
tphl=1.38276e-008 FROM 1.0005e-008 TO 2.38326e-008
tplh=9.84209e-009 FROM 8.0015e-008 TO 8.98571e-008
```

$$F_{max} \approx 84.5MHz$$

The CMOS inverter was successfully designed using Electric VLSI Design Tool. The layout consists of a PMOS transistor of width 2.5 um and an NMOS transistor of width

1 μm , with the input and output connections properly routed.

TABLE 0: CMOS Inverter Performance Comparison

Parameter	Without Capacitor	With Capacitor
Rise time t_r	20.8773 ns	21.3761 ns
Fall time t_f	24.9272 ns	25.5079 ns
t_{phl}	13.5094 ns	13.8276 ns
t_{plh}	9.62857 ns	9.84209 ns
Propagation delay t_{pd}	11.5690 ns	11.8348 ns

The results show that the extracted parasitic capacitance increases both the rise, fall times and propagation delay, reducing the maximum operating frequency. This agrees with the expected behavior of a practical CMOS circuit, since the additional parasitic capacitance must be charged and discharged during switching.

6 Conclusion

- CMOS inverter, 2/3-input NAND and NOR gates were successfully designed and simulated, and their switching characteristics were analyzed using rise time, fall time, and propagation delay.
- Transistor sizing was optimized to obtain approximately equal rise and fall times, while the effect of a 10 pF load capacitance on propagation delay and switching speed was also studied.
- The implemented inverter, NAND, and NOR gates form universal building blocks, allowing all other logic gates and more complex digital circuits to be constructed from them.